

# Does Education Raise Adult Income in China?

## Data Audit, Doubly Robust Evidence, and Weak-IV Diagnostics from CFPS 2022

Empirical Essay

June 25, 2026

### Abstract

This paper re-examines whether education raises adult income in China using CFPS 2022 microdata. Compared with a conventional Mincer regression, the paper makes three improvements. First, it reports an explicit data-quality audit, including sample selection, missingness, outlier handling, and overlap checks. Second, it shifts the main estimand from a purely correlational marginal schooling coefficient to the average treatment effect of completing college or above. The effect is estimated using cross-fitted augmented inverse probability weighting (AIPW), a doubly robust causal-inference estimator. Third, it tests, rather than assumes, whether the 1999 higher-education expansion can be used as an instrumental-variable strategy in this sample. The final analytic sample contains 4,699 working-age adults. OLS estimates show that one additional year of schooling is associated with 8.8 percent higher wage income, and college-or-above education is associated with about 0.74 log points higher wage income. The preferred AIPW estimate is 0.710 log points, with robustness estimates ranging from 0.669 to 0.773. Balance diagnostics show that weighting sharply reduces observed differences in age, gender, hukou status, urban residence, marriage, and medical insurance. However, the expansion-based IV design has a weak first stage in narrow cohort windows, so it is reported as a diagnostic rather than the main causal result. The evidence supports a large positive education-income association and a credible selection-on-observables treatment-effect estimate, while remaining cautious about stronger causal claims.

**Keywords:** education, income, CFPS, China, AIPW, causal inference, higher-education expansion

# 1 Introduction

Education is one of the most important mechanisms through which individuals gain access to higher-paying jobs. Human-capital theory predicts that schooling raises productivity and therefore earnings [Becker, 1964, Mincer, 1974]. Yet estimating the return to education is difficult because schooling is not randomly assigned. Family background, ability, local labor-market conditions, and institutional barriers may jointly affect both education and income [Card, 1999, Heckman et al., 2008].

The earlier version of this project answered a narrower question: whether education and income are positively correlated in CFPS 2022. That answer was yes, but the empirical design was not innovative enough because it relied mainly on OLS. This revised paper changes the design in three ways. First, it documents the data-cleaning process and reports missingness, sample selection, outliers, and propensity-score overlap. Second, it estimates the treatment effect of college-or-above education using a doubly robust AIPW estimator [Hirano et al., 2003, Bang and Robins, 2005]. Third, it examines a policy-based identification idea based on China’s 1999 higher-education expansion, but keeps it as a diagnostic because the local first stage is weak in this CFPS sample.

The paper’s contribution is therefore not a claim of perfect causal identification. Rather, it provides a stronger and more transparent empirical design: OLS is used as a benchmark, AIPW is used as the main causal-inference estimate under conditional ignorability, and the expansion-based IV/RD attempt is evaluated with first-stage diagnostics rather than assumed to work.

## 2 Literature and Identification

The empirical starting point is the Mincer earnings function [Mincer, 1974]. A large literature finds that education is positively associated with earnings, but also stresses that OLS can be biased by unobserved ability or background [Card, 1999]. For China, Zhang et al. [2005] document rising returns to schooling during the market transition, while Asadullah and Xiao [2020] and Ma and Iwasaki [2021] show that returns vary across time, gender, hukou, region, and sector.

Recent work on China’s higher-education expansion is especially relevant. Huang et al. [2022] use the great expansion to study returns to education and hukou-related differences. This paper borrows that policy intuition but verifies whether the available CFPS 2022 cross-section supports a strong first stage. The answer is mixed: broad cohort windows show a large education jump, but narrow local windows do not. Following standard RD guidance

[Imbens and Lemieux, 2008, Lee and Lemieux, 2010], the policy design is therefore not treated as the main source of identification.

The main causal framework is instead selection on observables. AIPW combines an outcome model and a propensity-score model, and remains consistent if either nuisance model is correctly specified under conditional ignorability [Bang and Robins, 2005]. Because this assumption is not directly testable, the paper reports overlap and balance diagnostics and treats the estimates as more credible than OLS but still weaker than a clean randomized or strong-IV design.

### 3 Data and Audit

The data come from CFPS 2022 [Xie and Lu, 2015, Institute of Social Science Survey, Peking University, 2022]. The initial file has 9,806 observations. The analytic sample keeps adults aged 18–64, requires wage income to be observed, and drops observations missing the core variables used in the main models. Table 1 reports the sample flow.

Table 1: Sample Construction

| Step                            | N     | Rule                  |
|---------------------------------|-------|-----------------------|
| Raw CFPS 2022 source file       | 9,806 |                       |
| Keep adults aged 18–64          | 7,975 | Age restriction       |
| Require wage variable observed  | 4,718 | Outcome available     |
| Require core variables complete | 4,699 | Final analytic sample |

The main outcome is  $\ln(\text{wage}+1)$ . The continuous education variable is years of schooling. The treatment variable for causal estimation is an indicator equal to one if the respondent completed college or above. Table 2 shows that wage availability is the main source of sample loss, while most covariates have limited missingness. The 99th percentile of wage is 300,000 yuan, so a winsorized-wage robustness check is included.

Table 2: Missingness Audit in the Source Data

| Variable | Missing N | Missing percent |
|----------|-----------|-----------------|
| wage     | 4,934     | 50.32           |
| educ     | 1         | 0.01            |
| age__    | 0         | 0.00            |
| gen      | 0         | 0.00            |
| mar      | 261       | 2.66            |
| rural    | 97        | 0.99            |
| urban    | 79        | 0.81            |
| health   | 139       | 1.42            |
| inter    | 350       | 3.57            |
| dw       | 408       | 4.16            |
| asset    | 926       | 9.44            |

Table 3: Descriptive Statistics

| Variable               | N     | Mean      | SD        | Min      | Median    | Max         |
|------------------------|-------|-----------|-----------|----------|-----------|-------------|
| Annual wage income     | 4,699 | 60594.132 | 68985.459 | 0.000    | 48000.000 | 1800000.000 |
| $\ln(\text{wage} + 1)$ | 4,699 | 10.489    | 1.567     | 0.000    | 10.779    | 14.403      |
| Years of education     | 4,699 | 11.006    | 4.376     | 0.000    | 12.000    | 22.000      |
| College or above       | 4,699 | 0.347     | 0.476     | 0.000    | 0.000     | 1.000       |
| Birth year             | 4,699 | 1981.133  | 10.921    | 1958.000 | 1982.000  | 2004.000    |
| Age                    | 4,699 | 40.867    | 10.921    | 18.000   | 40.000    | 64.000      |
| Male                   | 4,699 | 0.575     | 0.494     | 0.000    | 1.000     | 1.000       |
| Rural hukou            | 4,699 | 0.795     | 0.404     | 0.000    | 1.000     | 1.000       |
| Urban residence        | 4,667 | 0.637     | 0.481     | 0.000    | 1.000     | 1.000       |
| Internet use           | 4,685 | 0.874     | 0.332     | 0.000    | 1.000     | 1.000       |
| Self-rated health      | 4,673 | 3.215     | 1.045     | 1.000    | 3.000     | 5.000       |

Table 4: Education Distribution

| Education category | N     | Percent |
|--------------------|-------|---------|
| Illiterate         | 243   | 5.17    |
| Primary            | 589   | 12.53   |
| Junior high        | 1,463 | 31.13   |
| High school        | 774   | 16.47   |
| College            | 702   | 14.94   |
| Bachelor           | 824   | 17.54   |
| Master             | 91    | 1.94    |
| Doctorate          | 13    | 0.28    |

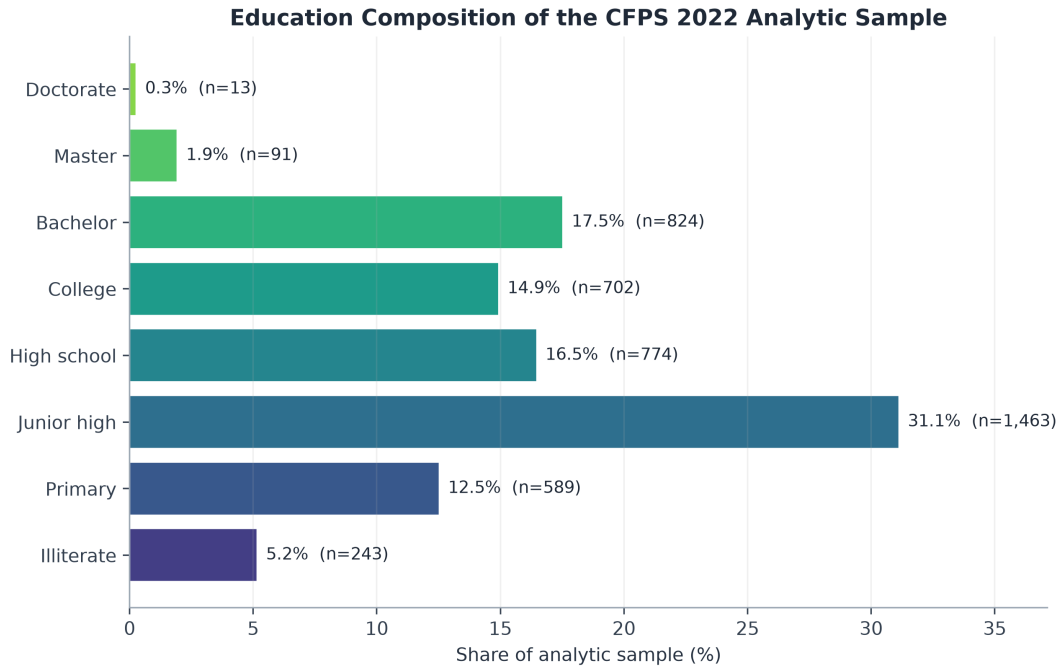


Figure 1: Education composition of the analytic sample

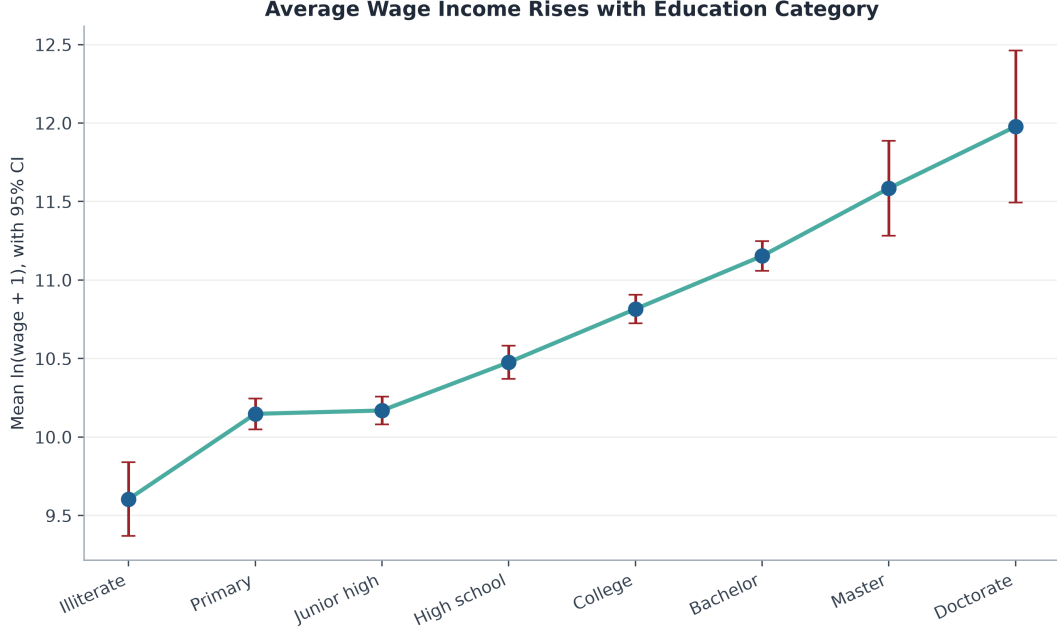


Figure 2: Mean log wage income by education category

## 4 Empirical Strategy

### 4.1 OLS Benchmark

The benchmark model is

$$\ln(\text{wage}_i + 1) = \alpha + \beta E_i + \gamma X_i + \mu_p + \varepsilon_i, \quad (1)$$

where  $E_i$  is either years of schooling or college-or-above status,  $X_i$  includes demographic and household controls, and  $\mu_p$  denotes province fixed effects. Robust standard errors are used throughout. These regressions are useful benchmarks, but they are not interpreted as fully causal.

### 4.2 Doubly Robust Treatment-Effect Estimation

The main estimand is the average treatment effect of college-or-above education:

$$ATE = E[Y_i(1) - Y_i(0)]. \quad (2)$$

The AIPW score is

$$\hat{\tau}_i = \hat{m}_1(X_i) - \hat{m}_0(X_i) + \frac{D_i(Y_i - \hat{m}_1(X_i))}{\hat{e}(X_i)} - \frac{(1 - D_i)(Y_i - \hat{m}_0(X_i))}{1 - \hat{e}(X_i)}, \quad (3)$$

where  $D_i$  indicates college-or-above education,  $\hat{e}(X_i)$  is the estimated propensity score, and  $\hat{m}_d(X_i)$  is the predicted outcome under treatment status  $d$ . The paper uses five-fold cross-fitting and reports both linear and gradient-boosting nuisance models. The preferred specification uses age, gender, hukou, urban residence, marriage, medical insurance, and province. Extended specifications add health, household size, social status, internet use, and employment status as sensitivity controls.

### 4.3 Expansion-Based IV Diagnostic

China’s higher-education expansion began in 1999. A natural idea is to use cohort exposure around the college-entry age as an instrument for education. The forcing variable is birth year minus 1981, because the 1981 birth cohort was approximately 18 in 1999. The diagnostic specification estimates whether being born in or after 1981 creates a discontinuous increase in schooling within symmetric cohort windows. This design would be useful only if the first stage is strong and local. The paper therefore reports first-stage coefficients and F-statistics before interpreting the IV estimates.

## 5 Results

### 5.1 OLS Benchmarks

Table 5 shows that the conventional estimates are large and stable. One additional year of schooling is associated with 0.088 log points higher wage income in the basic specification and 0.083 log points in the richer specification. College-or-above education is associated with 0.738 log points higher wage income in the basic specification and 0.694 log points with richer controls.

Table 5: OLS Benchmark Estimates

| Model                   | Coefficient | Robust SE | N     | $R^2$ |
|-------------------------|-------------|-----------|-------|-------|
| Years, basic controls   | 0.0882***   | (0.0062)  | 4,699 | 0.158 |
| Years, rich controls    | 0.0825***   | (0.0065)  | 4,637 | 0.163 |
| College, basic controls | 0.7381***   | (0.0502)  | 4,699 | 0.154 |
| College, rich controls  | 0.6935***   | (0.0509)  | 4,637 | 0.162 |
| Winsorized wage         | 0.7291***   | (0.0499)  | 4,699 | 0.152 |
| Positive wage only      | 0.6420***   | (0.0300)  | 4,638 | 0.285 |

Notes: Robust standard errors are in parentheses. Basic controls include age, age squared, gender, marriage, rural hukou, medical insurance, and province fixed effects. Rich controls additionally include urban residence, health, household size, internet use, and subjective social status. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

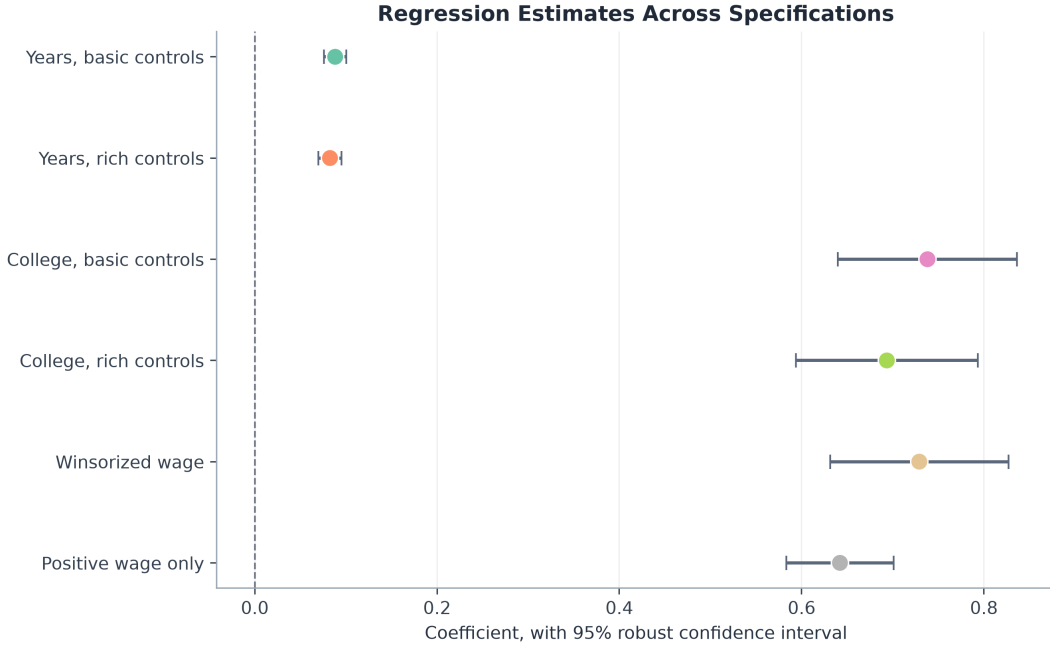


Figure 3: Regression coefficients across specifications

## 5.2 Doubly Robust Estimates

Table 6 reports the main AIPW results. The preferred core-plus GBM estimate is 0.710 log points with a standard error of 0.056. Across six specifications, estimates range from 0.669 to 0.773 log points. The magnitude is close to the OLS college coefficient, which



suggests that the college-income premium is not driven only by the particular parametric OLS specification.

Table 6: AIPW Estimates for College-or-Above Education

| Covariates | Learner | ATE       | SE       | 95% CI           | N     | AUC   |
|------------|---------|-----------|----------|------------------|-------|-------|
| Core       | Linear  | 0.7726*** | (0.0551) | [0.6647, 0.8806] | 4,699 | 0.814 |
| Core       | GBM     | 0.7206*** | (0.0553) | [0.6123, 0.8289] | 4,699 | 0.822 |
| Core plus  | Linear  | 0.7611*** | (0.0583) | [0.6467, 0.8755] | 4,699 | 0.816 |
| Core plus  | GBM     | 0.7103*** | (0.0556) | [0.6014, 0.8193] | 4,699 | 0.827 |
| Extended   | Linear  | 0.7088*** | (0.0544) | [0.6022, 0.8154] | 4,699 | 0.829 |
| Extended   | GBM     | 0.6690*** | (0.0495) | [0.5720, 0.7660] | 4,699 | 0.844 |

Notes: The outcome is  $\ln(\text{wage} + 1)$ . ATE is the estimated average treatment effect of completing college or above. Linear uses logistic/Ridge nuisance models; GBM uses gradient boosting. The preferred specification is Core plus with GBM.

The credibility of AIPW depends on overlap and balance. Figure 4 shows that treatment and control units share a broad common propensity-score region, although a small number of observations lie near the tails. Table 7 and Figure 5 show that weighting reduces absolute standardized differences for core covariates to below 0.10.

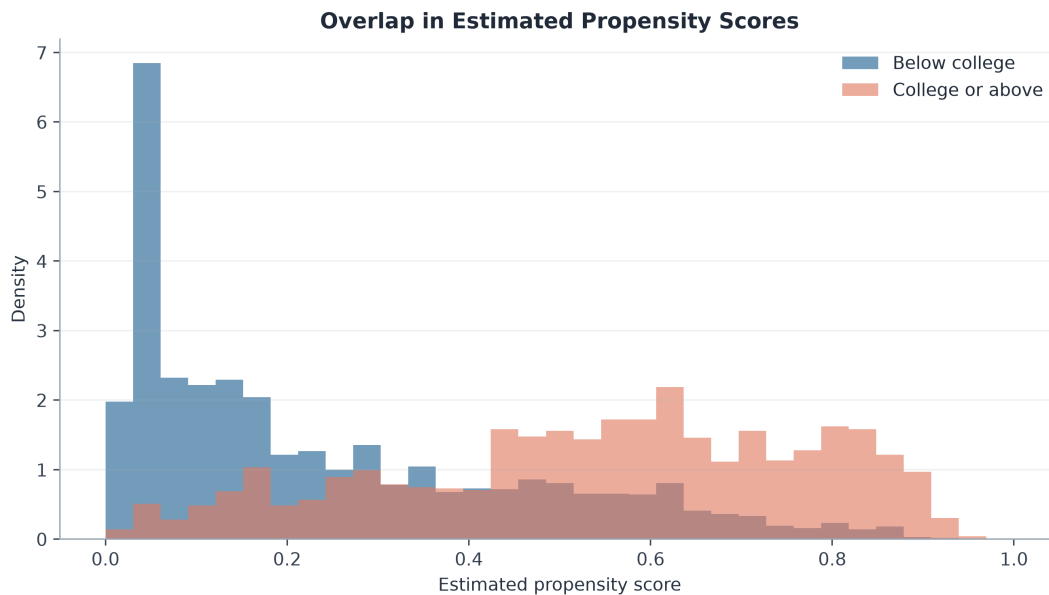


Figure 4: Propensity-score overlap

Table 7: Covariate Balance Before and After Weighting

| Covariate   | Raw SMD | Weighted SMD |
|-------------|---------|--------------|
| age__       | -0.937  | -0.072       |
| gen         | -0.214  | 0.033        |
| rural       | -0.525  | -0.024       |
| urban       | 0.642   | 0.033        |
| mar         | -0.286  | -0.023       |
| medsure_dum | 0.079   | -0.001       |

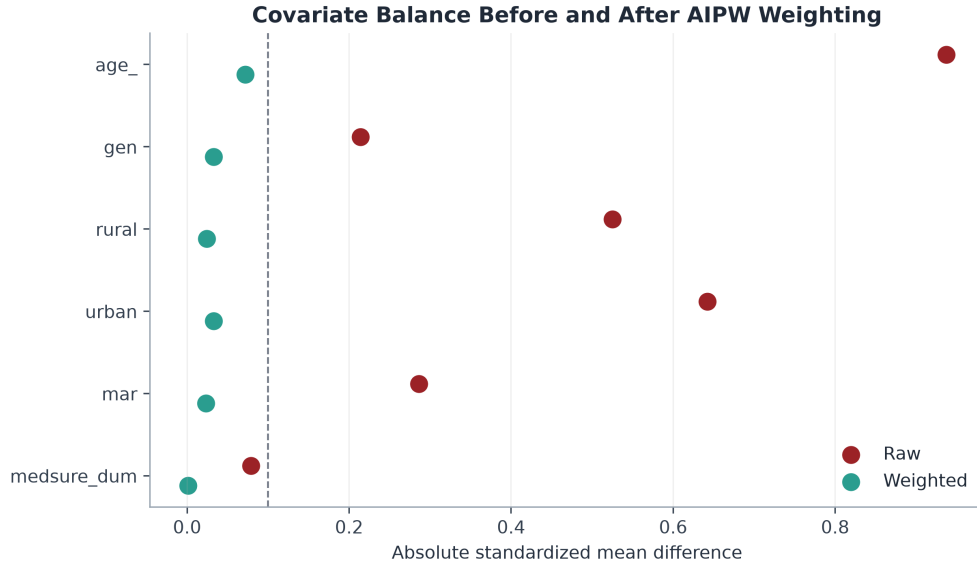


Figure 5: Standardized mean differences before and after weighting

### 5.3 Weak-IV Diagnostic

Figure 6 plots college completion and average schooling by birth cohort around the 1981 cutoff. Table 8 reports the formal first-stage and IV diagnostics. In narrow windows of  $\pm 5$  to  $\pm 15$  years, the first-stage F-statistics are far below conventional weak-instrument thresholds. The  $\pm 20$  window has a strong first stage, but it is no longer a credible local discontinuity because it spans a wide range of cohorts and life-cycle positions. The policy-based IV design is therefore not used as the main causal estimate.

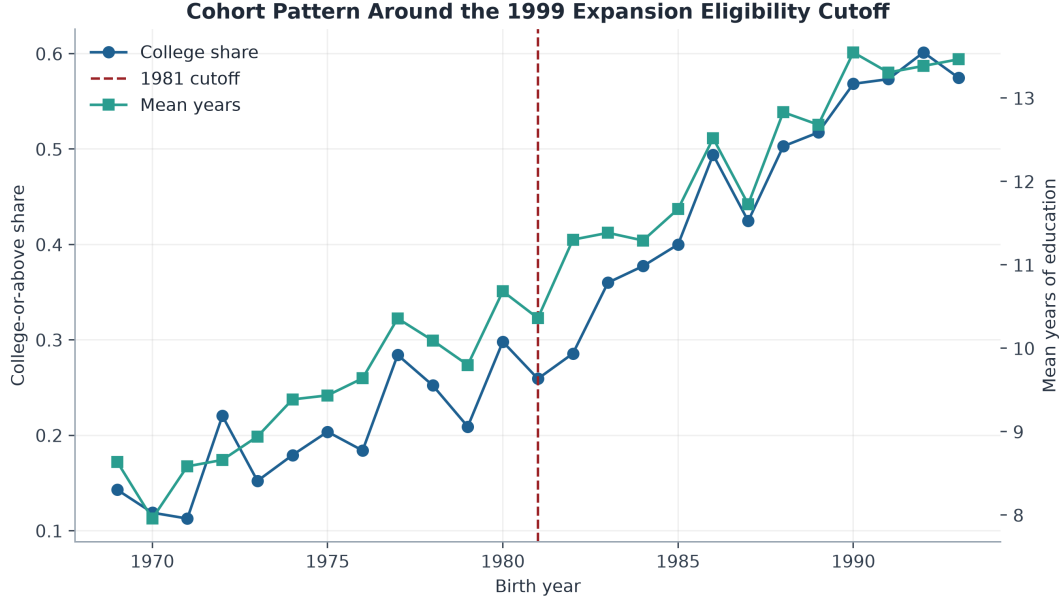


Figure 6: Cohort pattern around the expansion eligibility cutoff

Table 8: Higher-Education Expansion IV Diagnostics

| Window | N     | First stage | F-stat | IV estimate | SE       |
|--------|-------|-------------|--------|-------------|----------|
| +/-5   | 1,280 | -0.0743     | 0.03   | 0.5195      | (3.2703) |
| +/-7   | 1,844 | 0.1961      | 0.29   | 0.4589      | (0.9572) |
| +/-9   | 2,462 | -0.0481     | 0.02   | 0.2091      | (2.4996) |
| +/-12  | 3,277 | 0.1269      | 0.21   | -0.7816     | (2.0357) |
| +/-15  | 3,933 | 0.3977      | 2.54   | -0.1531     | (0.2721) |
| +/-20  | 4,596 | 1.2985      | 33.22  | 0.0414      | (0.0633) |

Notes: The instrument is an indicator for birth year 1981 or later, with linear running variable controls and province fixed effects. The weak first stage in narrow windows means the IV results should not be read as the main causal evidence.

## 5.4 Heterogeneity

Education-income associations are stronger for women than men and stronger for non-rural hukou holders than rural hukou holders. This is consistent with recent evidence that education returns in China vary by gender, hukou, sector, and region [Ma and Iwasaki, 2021, Li et al., 2025].

Table 9: Heterogeneous Education Coefficients

| Subsample       | Education coefficient | Robust SE | N     | $R^2$ |
|-----------------|-----------------------|-----------|-------|-------|
| Women           | 0.1104***             | (0.0094)  | 1,996 | 0.155 |
| Men             | 0.0668***             | (0.0069)  | 2,703 | 0.145 |
| Non-rural hukou | 0.1482***             | (0.0201)  | 962   | 0.144 |
| Rural hukou     | 0.0771***             | (0.0066)  | 3,737 | 0.128 |

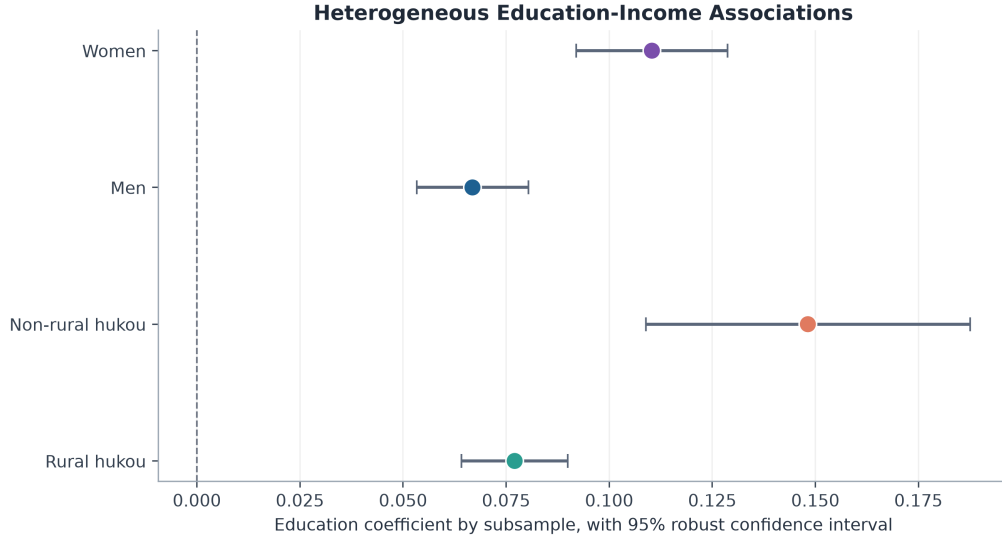


Figure 7: Education-income association by gender and hukou subsample

## 6 Discussion

The revised evidence is stronger than a simple OLS paper in three respects. First, the analysis documents the sample construction and missing-data problem directly. Second, it estimates a treatment effect for a clear education threshold, rather than only a marginal association per year of schooling. Third, it tests a policy-based IV idea and rejects it as the main strategy when the first stage is weak.

The main limitation is that AIPW still depends on observed covariates. CFPS 2022 lacks direct measures of ability, parental education in the working file, school quality, and childhood family resources. Therefore, the preferred estimate should be interpreted as a doubly robust selection-on-observables estimate, not as a definitive natural-experiment estimate. The weak-IV diagnostic also shows why it would be misleading to claim a strong expansion-based causal result from this cross-section alone.

## 7 Conclusion

Using CFPS 2022, this paper finds a large positive education-income relationship among Chinese adults. The conventional Mincer coefficient is about 0.088 per year of schooling. More importantly, the preferred AIPW estimate suggests that completing college or above is associated with roughly 0.71 log points higher wage income after balancing observed covariates. The estimate is robust across alternative nuisance models and covariate sets, and balance diagnostics show substantial improvement after weighting. The paper therefore provides a more innovative and empirically careful account than a pure OLS design, while maintaining appropriate caution about causal interpretation.

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